

ARDHI UNIVERSITY



FUNDAMENTAL OF COMPUTER NETWORKS.

INFORMATION SYSTEM MANAGEMENT (ISM 1).

GROUP ASSIGNMENT NO 11.

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NETWORK ACCESS LAYER.

Network Access Layer this is the fourth and last layer of the TCP/IP model , By its function it deals with getting packets from the source all the way to the destination. The Network Access Layer is more concerned with the set of all routers and links and compute appropriate paths through it, even for large networks as well explained in (*James.F.Kurose, 2025*) in its fifth chapter on first subtopic page no 359 up to 362.

It is responsible for the physical transmission of data and defines how bits are packaged into frames to move across a specific physical medium, It combines the functions of the Physical Layer and Data Link Layer of the OSI model , it combines some technology such as the ethernet and Wi-Fi to associate together enabling wired and wireless communication.

It provides the interface between the software-based protocols of the Internet Layer (like IP) and the physical hardware of the network, as its main function it performs the following;

- Encapsulation;-means that it takes the IP packets from the layer above and wraps into frames
- Media mapping;- It maps IP addresses to physical hardware addresses (MAC addresses).
- Physical Transmission;-It handles the electrical, optical, or radio signals used to move data.

"The Network Access Layer defines the protocols and hardware required to connect a host to a physical network and deliver data across it." — *Forouzan, B. A. (2017). Data Communications and Networking.*

KEY FUNCTIONS OF THE LAYERS.

The Network Access Layer translates software-based instructions into hardware reality through several critical tasks:

- **Framing:** It accepts IP packets from the Internet Layer above and encapsulates them into structural data units called frames. The frame includes hardware headers and trailers necessary for synchronization.

- **Physical (MAC) Addressing:** It uses unique, factory-assigned **Media Access Control (MAC) addresses** to reliably route frames between localized nodes on the same network segment.
- **Media Access Control (MAC Sublayer):** It enforces protocols like CSMA/CD (for wired networks) or CSMA/CA (for wireless networks) to manage how devices share physical media and prevent data collisions.
- **Error Detection:** It appends a Frame Check Sequence (FCS) to verify the data integrity of received frames, discarding corrupted data at the hardware level.
- **Signal Conversion:** It transforms binary data (bits) into physical signals (e.g., electrical currents, light pulses, or radio waves) suited for the specific medium

1. PHYSICAL TRANSMISSION /THE CONVERSION BITS (The "Signal").

This part deals with the actual movement of data. It defines how bits (1s and 0s) are represented by physical changes in a medium, or it may also be defined as is the process of converting a frame into a stream of signals (electrical, optical, or radio) that can travel across a specific medium. This involves line coding or modulation, where the binary 1s and 0s are mapped to physical states, such as voltage levels or light pulses.

- **Copper Cables (Ethernet):** Uses variations in electrical voltage.
- **Fiber Optics:** Uses pulses of light.
- **Wireless (Wi-Fi):** Uses changes in radio frequencies.

Some books and resources explain in deep about the case of physical transmission such as;

"Physical transmission is the mechanical, electrical, and functional means to activate, maintain, and de-activate physical-level connections for bit transmission between data-link entities." — Stallings, W. (2016). *Data and Computer Communications*.

2. HARDWARE ADDRESSING (The "Name Tag")

Every device has a permanent, unique "name tag" called a **MAC Address** (Media Access Control). While an IP address is like a mailing address that changes if you move, a MAC address is like your Social Security number—it stays with the hardware forever. This means that Hardware Addressing (specifically MAC addressing) is used for delivery within a single local network segment. Every Network Interface Card (NIC) is assigned a unique 48-bit address, which the Network Access Layer uses to ensure that a frame reaches the specific physical port of the intended recipient.

- **The Process:** Before data leaves your computer, the Network Access Layer attaches the MAC address of the destination device to the data.
- **The Purpose:** This ensures that in a room full of computers, only the intended recipient picks up the data "frame."

"Hardware addressing, or MAC addressing, provides the unique identification required for devices to communicate on a shared local medium, ensuring that frames are delivered to the correct physical destination regardless of the logical network structure." — Kurose, J. F., & Ross, K. W. (2021). Computer Networking: A Top-Down Approach.

This means while routing packets, before the physical transmission the hardware creates the MAC addressing for its device ensuring the receiver to get the address and to ensure the packet goes to the right place.

"The Media Access Control sublayer is responsible for determining which device has the right to use the physical medium at any given time, preventing signal interference and data loss." — Tanenbaum, A. S., & Wetherall, D. J. (2011). Computer Networks.

COMMON PROTOCOLS AND NETWORK TECHNOLOGIES.

The Network Access Layer is diverse because it must support various types of physical media (copper wires, fiber optics, and airwaves).

**THE TABLE FOR SOME COMMON PROTOCOLS AND HARDWARE COMPONENT IN
NETWORK ACCESSED LAYER.**

Technology	Medium	Standard Reference	Specification
Ethernet	Twisted Pair/ Fiber	IEEE 802.3	For wired local Area Networks using copper or fiber-optic cinnections
Wi-Fi	Radio waves	IEEE 802.11	Standard architecture for wireless local networks
Point to Point	Serial/Fiber Links	RFC 1661	Direct communication between two dedicated networks
ARP	Address Mapping	RFC 826	Mapping a logical IP address to a local physical MAC address.

References;

Forouzn, B. (2017). *Data Communication and Networking*. New York: Global Edition.

James.F.Kurose. (2025). *Computer Networking*. Chicago,IL: Pearson.

Stallings, W. (2016). *Data and Computer Communications*. Pearson.

Tanenbaum,A.S.,& Wetherall,D.J. (2011). *Computer Networks*.

